

ATS Resume Scoring — Research Report

Task 1 — Week 2, Resume AI Project

1. What is an ATS?

An Applicant Tracking System (ATS) is software that companies use to manage job applications. When a candidate applies for a job, their resume usually does not go straight to a human recruiter. Instead, it goes into the ATS first. The ATS stores the resume, reads its content, and helps the recruiter filter, search, and rank candidates before anyone manually looks at them.

Large companies receive hundreds or even thousands of applications for a single job opening. It would take too long for a human to read every resume, so the ATS acts like a first filter — narrowing down the pile to the resumes that seem most relevant to the job.

2. How Does ATS Rank Resumes?

Most ATS platforms rank resumes by comparing the text of the resume against the text of the job description. The system looks for overlap — words, phrases, skills, and qualifications that appear in both documents. The more a resume matches what the job description is asking for, the higher it is ranked.

Typically, the ranking process works like this:

- The resume is uploaded and converted into plain text (this is called parsing).
- The system extracts key information: skills, job titles, years of experience, education.
- It compares this extracted information to the requirements listed in the job description.
- A score is calculated, usually as a percentage, showing how closely the resume matches.
- Resumes are sorted from highest to lowest score, and recruiters usually look at the top ones first.

3. What Factors Influence ATS Scores?

Several things affect how well a resume scores in an ATS:

- Keyword match — does the resume contain the specific words used in the job description (e.g. 'Python', 'project management')?
- Job title match — does the candidate's past job title match or relate to the title being hired for?
- Skills section — a clear, well-labeled skills section is easier for the ATS to read and match.
- Formatting — simple, clean formatting (no tables, columns, images, or unusual fonts) is easier for the ATS to parse correctly.
- Years of experience — some systems check if the candidate meets a minimum experience requirement.
- Education and certifications — degrees or certifications mentioned in the job posting.
- File type — .docx and simple PDF files are usually read more reliably than scanned images or heavily designed PDFs.

4. Keyword Matching vs Semantic Matching

Keyword matching is the older, simpler method. The ATS simply checks whether an exact word or phrase from the job description also appears in the resume. For example, if the job description says 'customer service' and the resume says 'client support' instead, a keyword-matching system may not recognize these as related — even though they

mean almost the same thing. This is a major limitation, because it rewards resumes that copy exact wording rather than resumes that show real matching experience.

Semantic matching is a newer, smarter approach that uses AI to understand meaning rather than just matching exact words. Using techniques like sentence embeddings (explained further in Task 2), a semantic system can recognize that 'client support' and 'customer service' are related concepts, even though the words are different. This allows for a fairer and more accurate comparison between a resume and a job description, because it evaluates meaning instead of just exact text.

5. Common Mistakes That Reduce ATS Scores

- Using images, tables, or columns — many ATS systems cannot read text placed inside these elements.
- Using headers/footers for important information — some systems skip over headers and footers entirely.
- Uncommon section titles — e.g. naming the skills section 'What I Bring' instead of 'Skills' can confuse parsing.
- Missing keywords from the job posting — if the resume never mentions terms the job description uses, the match score drops.
- Overly creative fonts or designs — unusual fonts or graphics-heavy templates can be misread or skipped entirely.
- Saving as an image-based PDF (like a scanned document) — the ATS cannot extract text from an image.
- Inconsistent formatting of dates, job titles, or section headers — this can confuse automatic field extraction.

6. How AI Can Improve Resume Evaluation

Traditional ATS systems rely heavily on exact keyword matching, which can unfairly reject good candidates simply because they used different wording. AI-based evaluation improves this process in a few important ways:

- Understanding meaning, not just words — AI models can recognize that different phrases can express the same skill or experience.
- Ranking by true relevance — instead of counting keyword occurrences, AI can measure how closely the overall meaning of a resume matches a job description.
- Identifying skill gaps — AI can highlight which important skills from the job description are missing from the resume, and suggest what a candidate could add or learn.
- Reducing bias from exact-wording mismatches — semantic understanding helps ensure that strong candidates are not filtered out purely due to different phrasing.
- Providing feedback — instead of a system that just accepts or rejects, an AI-powered tool can explain the score and offer a roadmap for improvement.

This is the direction this project is heading in: moving beyond simple keyword counting toward a system that understands resumes and job descriptions the way a human recruiter would, but faster and more consistently.